

Modeling and Predicting the Behaviors of Inductive Traffic Loops

Cooper Amador-Risch, Tushar Mehta, Swetha Menon

Abstract

Inductive Traffic Loops are a traffic control technology that aid in the facilitation of roadways and data collection for traffic systems. These loops are used throughout the world, but are not fully utilized to gain all the data possible from traffic systems. While cameras are becoming increasingly prevalent across the globe, many areas still use inductive traffic loops. These areas should still be collecting valuable traffic data in order to improve traffic control. We hypothesized that the inter-lane position of motorcycles could be predicted using the inductance values of the traffic loop. To prove this, we started with Ansys HFSS simulations of inductive traffic loops with motorcycles passing over them to find a pattern of inductance changing across different inter-lane positions. To generate larger amounts of data, we generated random types of motorcycles by using Gaussian Distribution Noise and we generated these motorcycles' change in inductance values at each position by using Hyperbolic Secant Distribution Noise. We also utilized a Physics Informed Neural Network (PINN) in order to detect the inter-lane position for any of these motorcycles. This network utilized physics equations and classifications in order to make better predictions about the position a motorcycle was likely to be in. While people believe that inductive traffic loops are becoming more outdated, our findings indicate that they can still provide detailed, useful information about the positioning of vehicles on roadways through the utilization of unique and innovative AI and Machine Learning algorithms.

I. Introduction

A. Relevance

Modern transport systems use smart infrastructure tools to help mitigate congestion and increase smooth traffic flow. Among the most widely adopted tools in infrastructure planning are inductive traffic loop detectors; electromagnetic sensors embedded in roadways that detect the presence of vehicles. The detectors produce data for intelligent transport systems, enabling applications such as dynamic signal control, congestion monitoring, and traffic management. Inductive traffic looping refers to the strategic placement and use of inductive loops to monitor traffic at critical junctions and road segments.¹ When a vehicle passes above the loop, it induces a change in inductance. This induced change is detected by roadside electronics and translated into data.

B. Electromagnetic Theory

Inductive traffic loops use electric and magnetic fields to detect the presence of vehicles above the road. When an alternating current flows through a loop of wire embedded beneath the roadway, it generates an oscillating magnetic field that projects vertically above the pavement, known as the detection zone. As a vehicle enters this zone, its conductive and ferromagnetic components disrupt the surrounding field by generating eddy currents or by local magnetic permeability (how much magnetic flux a material supports). These interactions change the loop's inductance, which in turn affects the circuit's resonance frequency.

The loop detector circuit is tuned to a specific resonant frequency, and the change in inductance affects the resonant frequency of the circuit, and this change signifies a vehicle's presence. The process of traffic loop inductance is governed by the equations seen below.

$$\oint B \cdot dl = \mu_0 I_n \quad (1)$$

Ampère's Law (Eq.1) states that electric currents generate magnetic fields that circulate around the path of the current. A current carrying wire produces a magnetic field that forms concentric loops around the wire, and its strength depends on the magnitude of the current and radial distance from the wire. The line integral of the magnetic field along a closed path, an Amperian loop, is given by the equation above. The line integral of the magnetic field vector multiplied by the infinitesimal path element dl , or dr^* , is equal to the permeability of free space μ_0 multiplied by the net electric current passing through the area surrounded by the Amperian loop I_0 .

$$\oint E \cdot dA = \frac{Q}{\epsilon_0} \quad (2)$$

Gauss's Law for Electricity (Eq.2) states electric fields originate from electric charges, and the strength of those fields depends on the amount and distribution of charge within a closed surface. The equation describes how the total electric flux through a closed surface is proportional to the net charge enclosed within the surface. E represents the strength and direction of the electric field, dA represents the infinitesimal area vector, Q represents the total electric charge in the closed surface, and ϵ_0 is a permittivity for free space constant. E represents the strength and direction of the electric field, dA represents the infinitesimal area vector, Q represents the total electric charge in the closed surface, and ϵ_0 is a permittivity for free space constant.

$$\oint B \cdot dA = 0 \quad (3)$$

Gauss's Law for Magnetism (Eq.3) states that a net magnetic flux through any closed surface is equal to zero. This reflects the fact that magnetic monopoles do not exist, as magnetic field lines form loops – there is no isolated north or south pole where lines start or stop.²

Faraday's Law for Electromagnetic Induction, the fourth Maxwell equation will be explained in section C.

C. Induction

Electromagnetic induction is the process by which a changing magnetic field induces an electromotive (EMF) force in a conductor. This phenomenon was discovered by Michael Faraday in 1831 and later formalized by James Maxwell in part of his unified theory of electromagnetism.

$$\phi_B = BA \cos(\theta) \quad (4)$$

Above is the magnetic flux equation (Eq.4), created by Michael Faraday who discovered a changing magnetic flux through a coil induces an electromotive (EMF) force, which can drive a current. Flux, by definition, is the measure of the total magnetic field that passes through a surface. This discovery laid the

foundation for electric generators, transformers, and inductive sensors. Magnetic flux (ϕ_B) is equal to the strength of the magnetic field (B), the area through which the field passes (A), and the angle between the magnetic field and the surface (θ).³

$$\epsilon = -\frac{d\phi_B}{dt} \quad (5)$$

The magnetic flux is directly related to Faraday's Law of Electromagnetic Induction (Eq.5). Faraday noticed only changing magnetic fields could induce currents, and that static fields had no effect. ϵ is the

induced electromagnetic force in volts, $\frac{d\phi_B}{dt}$ is the derivative of magnetic flux with respect to time. The negative sign comes from Lenz's Law, indicating that the induced EMF opposes the change in the flux—as the magnetic flux increases, the induced current creates a magnetic field that opposes the increase. This opposition ensures the system resists sudden changes, similarly to inertia in mechanics.

$$\epsilon = -L\frac{dI}{dt} \quad (6)$$

Self inductance (Eq.6) is a specific case of Faraday's Law where the changing magnetic flux is caused by the coil's own changing current. Coils experiencing self inductance generate an electromotive force that opposes changes in the current flowing through them. This phenomenon is explained by Lenz's Law,⁴ which states that an induced EMF resists the change that created it. The strength of the self-induced electromagnetic force depends on the coils self-inductance, a property determined by its

geometry, number of turns, and core material. L represents the self inductance in henries, and $\frac{dI}{dt}$ represents the rate of change of the current.

$$\epsilon_2 = -M\frac{dI_1}{dt} \quad (7)$$

Mutual Inductance (Eq.7) is where a changing current in one coil induces an electromotive force in another coil due to the shared magnetic field between them. When a current flows through the first coil, it generates a magnetic field that partially links with the second coil. If the current in the first coil varies over time, the changing magnetic flux through the second coil induces a voltage across it. The strength of this interaction is quantified by the mutual insurance M , which depends on factors including the number of turns in each coil, orientation, distance, and magnetic permeability of the medium in between them. ϵ_2 represents the induced electromotive force in coil due to coil 1 with I_1 being the current in coil 1. ⁵

Inductive traffic loops use self insurance rather than mutual inductance because the detection mechanism depends on how the loop reacts to changes in its own magnetic field, not on interaction between separate coils. The loop senses alterations in self inductance caused by conductive vehicles overhead, which distort and weaken the magnetic flux within the loop itself.

D. Ferromagnetic Effects and Magnetic Movements

Ferromagnetism is due to the collective alignment of atomic magnetic moments, or the magnetic dipoles generated by the spin and orbital motion of unpaired electrons. In ferromagnetic materials such as iron,

cobalt, and nickel, the moments align in the same direction within regions called magnetic domains, even without an external magnetic field. This collective alignment is driven by a quantum mechanical phenomenon known as exchange interaction, which energetically favors a parallel spin orientation among neighboring atoms. As a result of the spin, the material exhibits a strong net magnetization. When an external magnetic field is applied, these domains grow and orient themselves in a way that amplifies the magnetic response.

In traffic loops, the vehicles' ferromagnetic components, such as the engine, alter the magnetic field generated by the loop. The exchange interaction enables ferromagnetism, which gives vehicles the magnetic properties that make inductive detection possible. ⁶

E. Loop Circuit

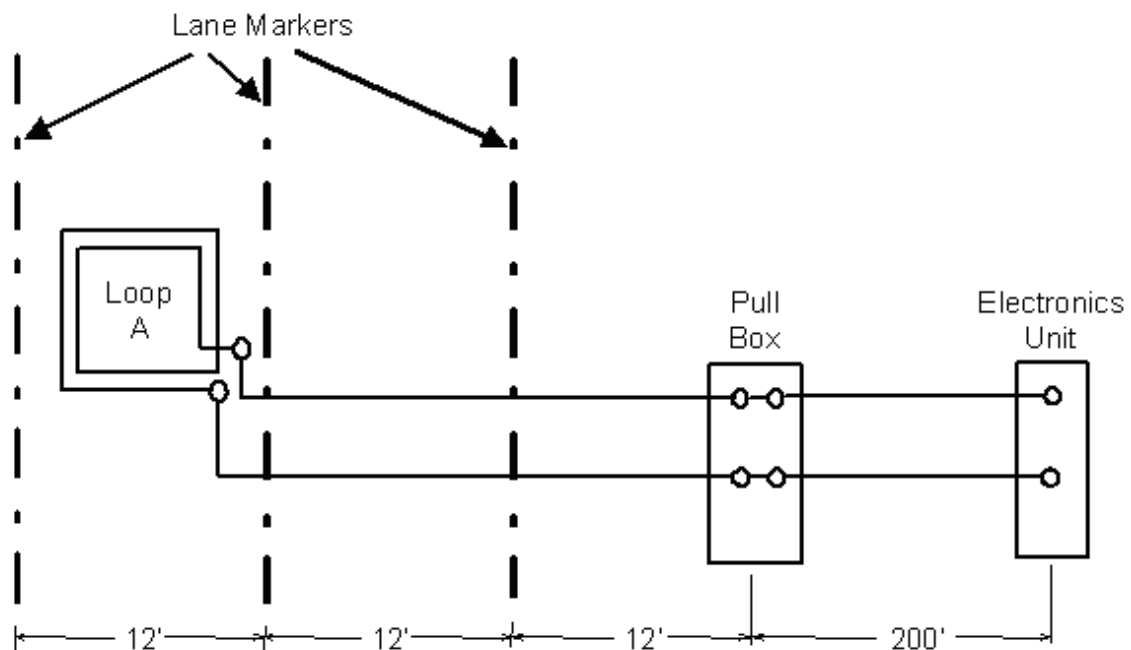


Figure 1: Inductive Traffic Loop Circuit¹

The principal components of an inductive loop system include one or more turns of an insulated loop wire in the pavement, a lead in cable from the curbside pull box to the intersection controller cabinet, and an electronics unit in a nearby controller cabinet (see Figure 1). More specifically, the electronics unit transmits an alternating current to the wire loop through the lead-in cable, typically between 10 kHz and 200 kHz. This creates an oscillating magnetic field above the loop called a detection zone. When a vehicle passes over the loop, the vehicle induces eddy currents, which lead to the decrease in the inductance of the wire. The change in inductance actuates an output relay which sends a pulse to the controller to signify the passing of a vehicle. ¹

F. Traffic Theory

Traffic theory is the study of how vehicles move through transportation networks, focusing on flow (vehicles per hour), density (vehicle per mile), and speed (miles per hour). Traffic consists of free flow and congested traffic. The three phase traffic theory by Boris Kenner states that there are three different phases: free flow, synchronized flow, and wide moving jam.

Free flow is when traffic moves at high speed with minimal interaction. The vehicles are spaced out meaning drivers will be able to travel at the speed limit without frequent breaking or lane changes. Synchronized flow is when the speed drops, but the vehicles continue moving; drivers in different lanes tend to match each other's speed, and traffic congestion stays near bottlenecks and does not move forward. Wide moving jams are when vehicles are nearly or are stationary inside the jam. These jams maintain their shape and speed as they move upstream even if bottlenecks appear. Kerner's traffic theory emphasizes that wide moving jams do not come directly from free flow but progress from free flow to synchronized flow to a wide moving jam.

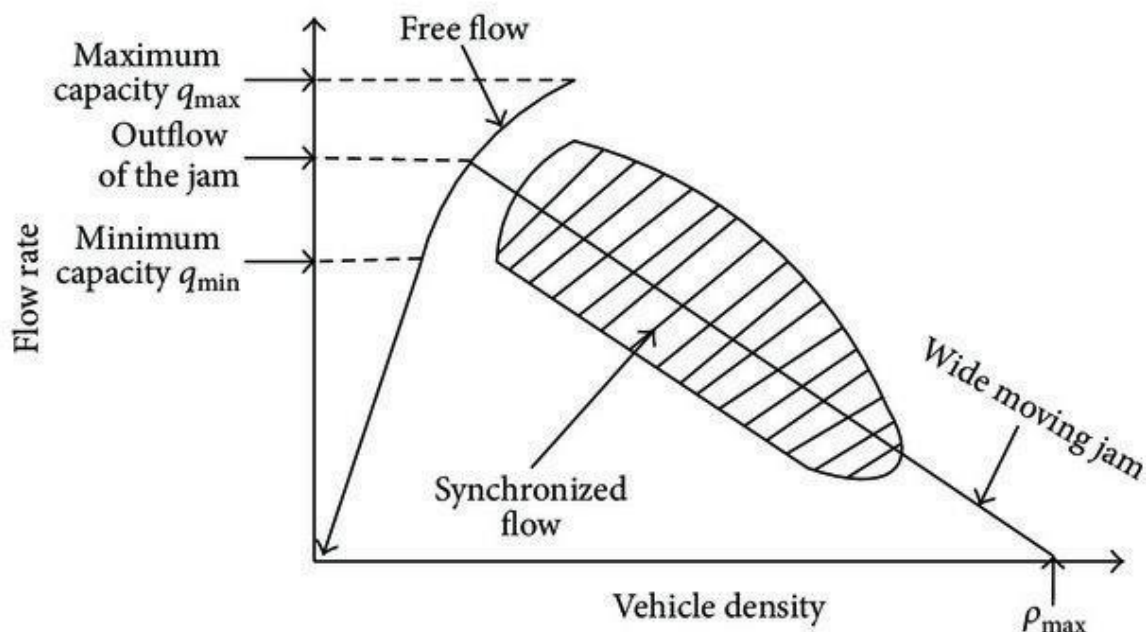


Figure 2: Boris Kenner's 3 Phase Traffic Theory Graph⁷

The graph (see Figure 2) shows how traffic flow varies with vehicle density, illustrating the three distinct phases. In the free flow region, the vehicles are spaced far apart and move quickly, so the flow increases with density. As the density increases, traffic shifts to the synchronized flow phase, where the speed drops and vehicles across lanes begin to match each other's pace. This region is broad because there is variability of synchronized traffic. At the higher densities of traffic, synchronized flow turns into a wide moving jam, where vehicles are almost stationary. ⁷

Even though this theory focused on highway systems, the same principals can be applied to any roadway system. Just as highways on ramps can cause bottlenecks, inefficient traffic systems can cause

congestion and bottlenecks. Principals of traffic theory can help systems anticipate congestion, preemptively reroute to maintain free flow, and adjust signal timing to dissolve jams.

G. Using Ansys To Model

Ansys Electronics Desktop is a simulation software that has many computational packages within it. We utilized the HFSS environment which allows us to simulate electric and magnetic fields at different operating frequencies. HFSS utilizes Finite Element Method (FEM) to solve the physical setup defined by the user. There is a built-in Computer Aided Design (CAD) modeler that allows for the 3D objects to be created with specified materials. HFSS also has numerous algorithms that allow for multifunctional analysis. The Parametric and Frequency sweeps allow for numerous design variables such as lengths and frequencies to be tested in a single analysis run. HFSS also has an optimization solver called Optimetrics which allows for certain design criteria to be optimized based on user-defined constraints. The simulation done for this project was done fully in HFSS.

H. Project Goal

The goal of this project is to model and simulate an inductive traffic loop, and to detect the presence of a motorcycle over the traffic loop.

II. Simulation

An Ansys HFSS simulation was done to gather the data about the behavior of inductive traffic loops when interacting with vehicles. Simulation requires specific setup steps in order for it to acquire accurate data.

A. Inductive Loop Setup

1. Modeling the Loop

Inductive traffic loops have general designs, but multiple possible geometries (see Figure 3) were considered.

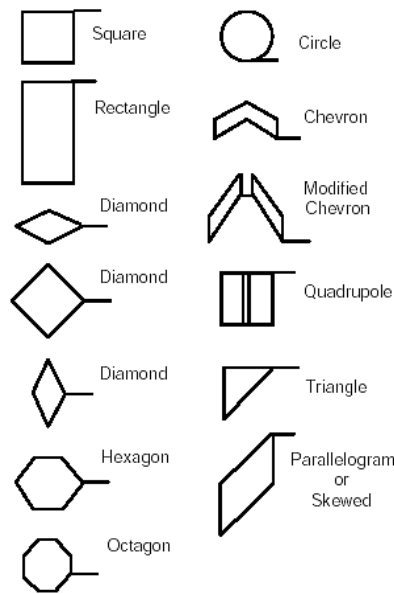


Figure 3: Inductive Traffic Loop Designs

Rectangular Loops are ideal at intersections and gates and can provide greater detection area, but do not provide a uniform detection field. Quadrupole Loops are great at reducing external noise from other lanes in multi-lane traffic systems, but have more complex geometries and are less common. Circular Loops have also been theorized as being more efficient but difficult installation and lacking widespread use make them less realistic. We decided to use the Square Loop design because it provides uniform sensitivity in all directions and is very common in most traffic systems. ¹

Ansys HFSS has modelling tools that were utilized in order to create the square loop. In coordination with parametric design, we ensured that all sides of the loop were variable driven in case needed changes in the design arose. To create the actual wire 3D geometry, we swept a face around the edge of the line dimension (see Figure 4). We also assigned a copper material to the wire so that it behaved as copper wire.

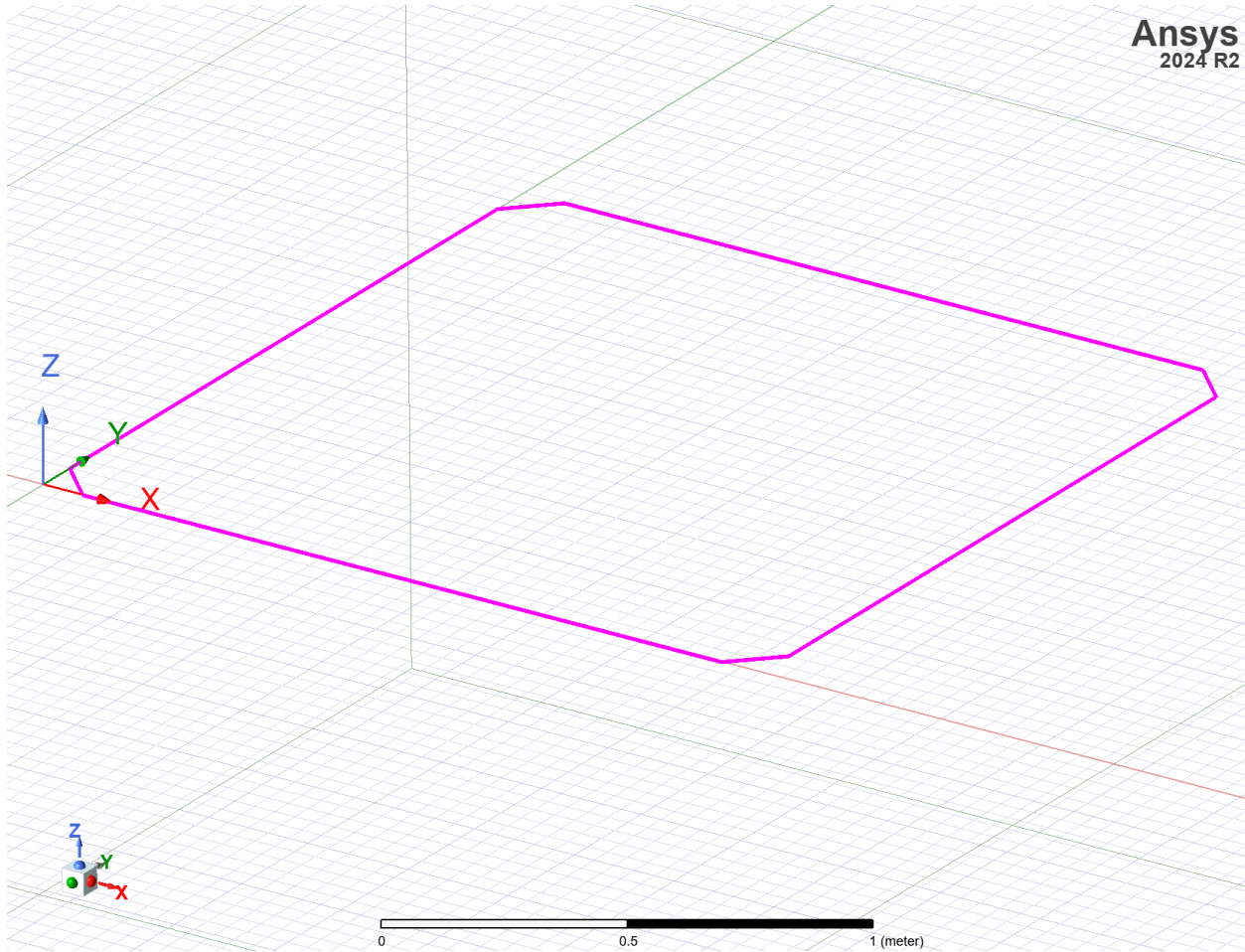


Figure 4: Modeled Square Inductive Traffic Loop

2. Boundary Conditions

Boundary Conditions refer to the regions which define the areas of simulation. The main region encapsulated all objects of the simulation. It was made an open boundary so all EMF fields could propagate out of the model volumes. This region was also defined as the Mesh Region, meaning all objects were meshed and simulated in this region (see Meshing sections). This boundary was defined as air material.

A sub region was created which was used to act as the concrete the loop is embedded in. This region is important because it restricts the loop's ability to propagate magnetic fields.

3. Excitation

An excitation of AC voltage needs to be applied to the loop in order to create the electric and magnetic fields that interact with the vehicles. To do this, HFSS uses Ports. Ports are used to create excitations across electric fields. We utilized the lumped port on the top face of the loop as that is where the electric field is (see Figure 5).⁸

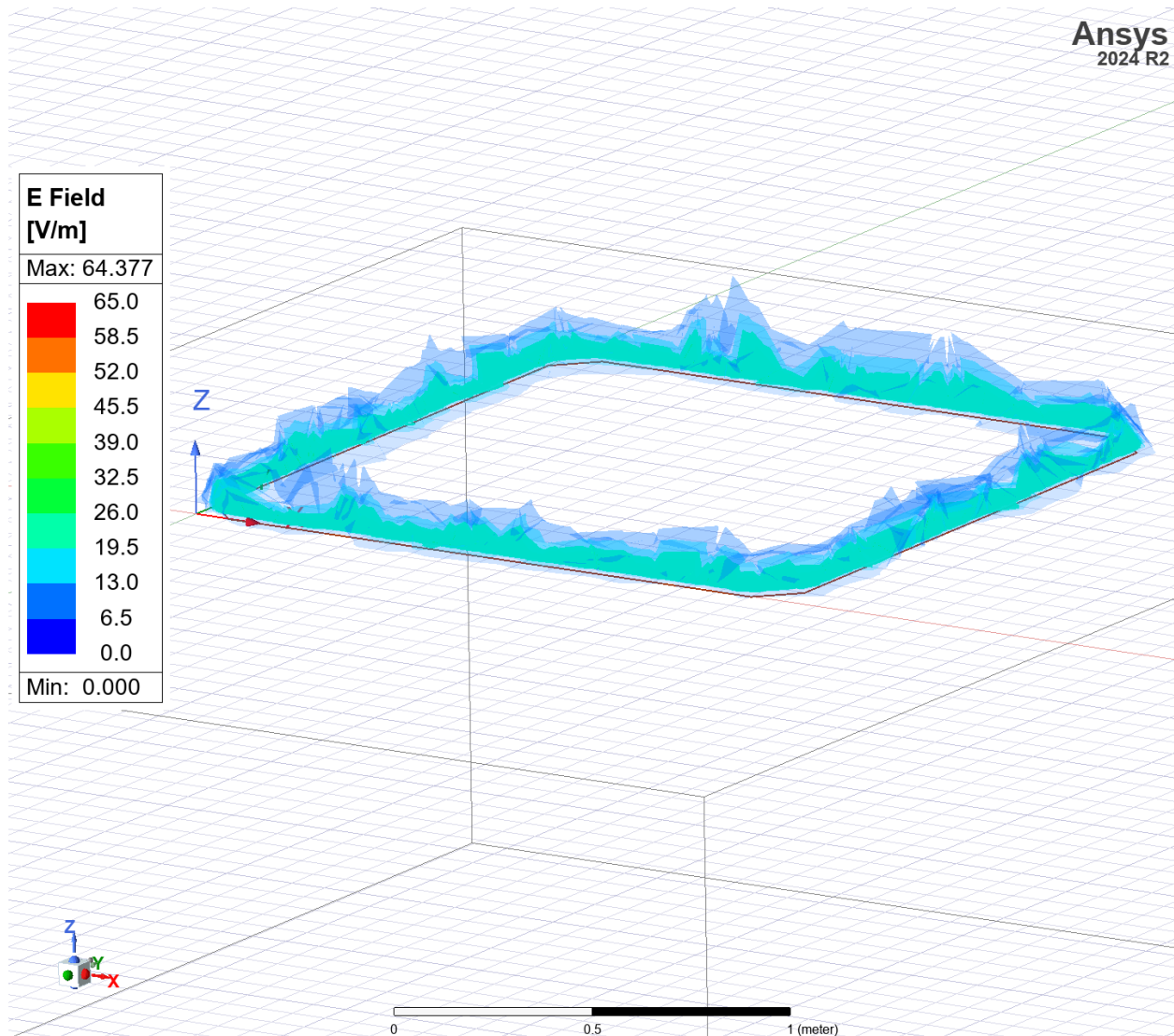


Figure 5: Electric Field of the Inductive Traffic Loop

4. Meshing

Meshing is at the core of FEM (see sec. 4 in Solver Setup). In short terms, the mesh is used to calculate the physics of the simulation. In HFSS, Mesh Operations are used in order to define how the meshing should be defined on different objects within the main simulation region. A mesh was defined on the loop with a maximum length of 0.3 meters in order to ensure accuracy and efficiency.

B. Vehicle Setup

1. Modeling the Vehicle

The vehicle was modeled as a box in order to keep the complexity of our simulations down. This box was parameterized using project variables in order to easily change the size of the box to model different vehicles. The box also had a parameterized position which allowed it to be easily moved to different locations adjacent to the loop. The vehicle material was set to cast iron.

2. Meshing

The vehicle was meshed with a surface mesh driven by length (see Figure 6). Used a medium mesh size to balance accuracy and speed of analysis.

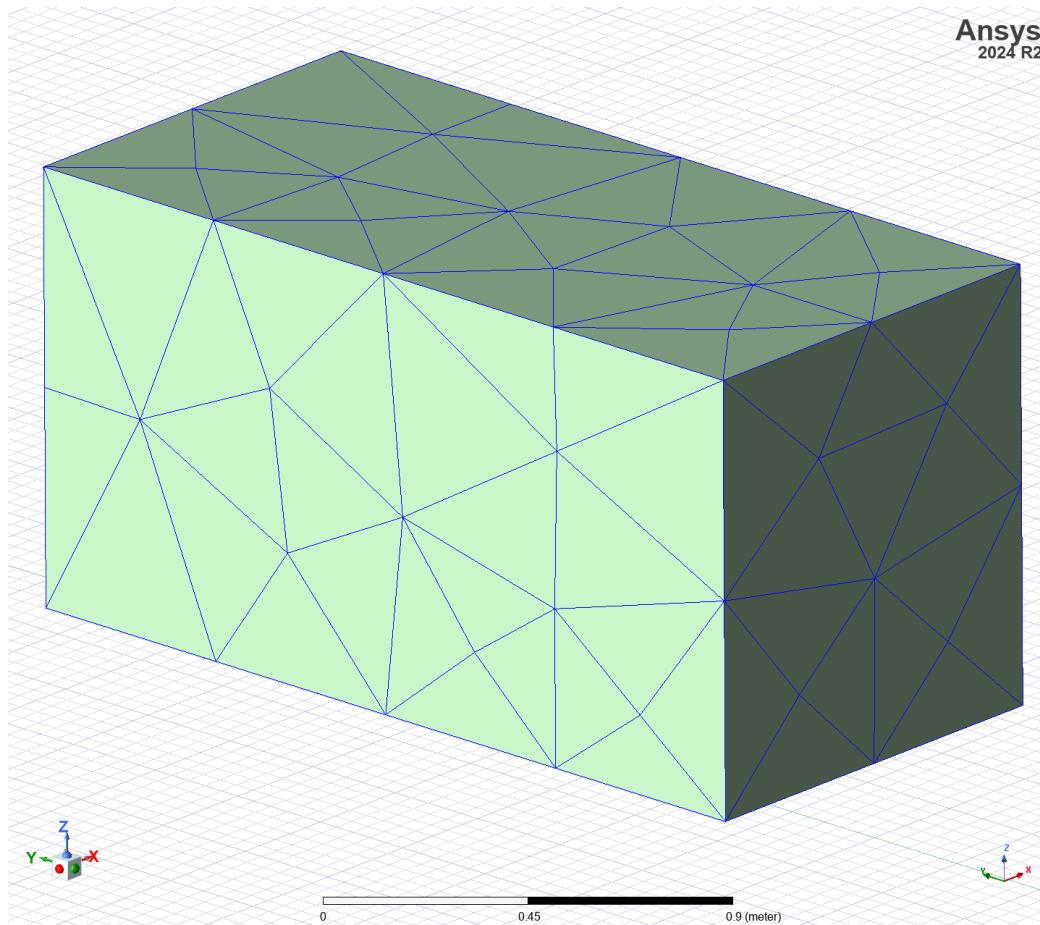


Figure 6: Vehicle Model

C. Solver Setup

The solver for HFSS requires certain parameters to be defined by the user. In our case, we utilized the default settings as those work for most scenarios including ours.

1. Frequency Sweep

In the solution setup, we defined a frequency sweep for the analysis. This essentially means that the model is tested at different operating frequencies in order to determine how it behaves across them. These frequencies are applied to the port that is on the loop for excitation. 1 kHz to 20 kHz was the range of frequencies tested. Our models ended up having similar behavior across frequencies (see Figure 7).

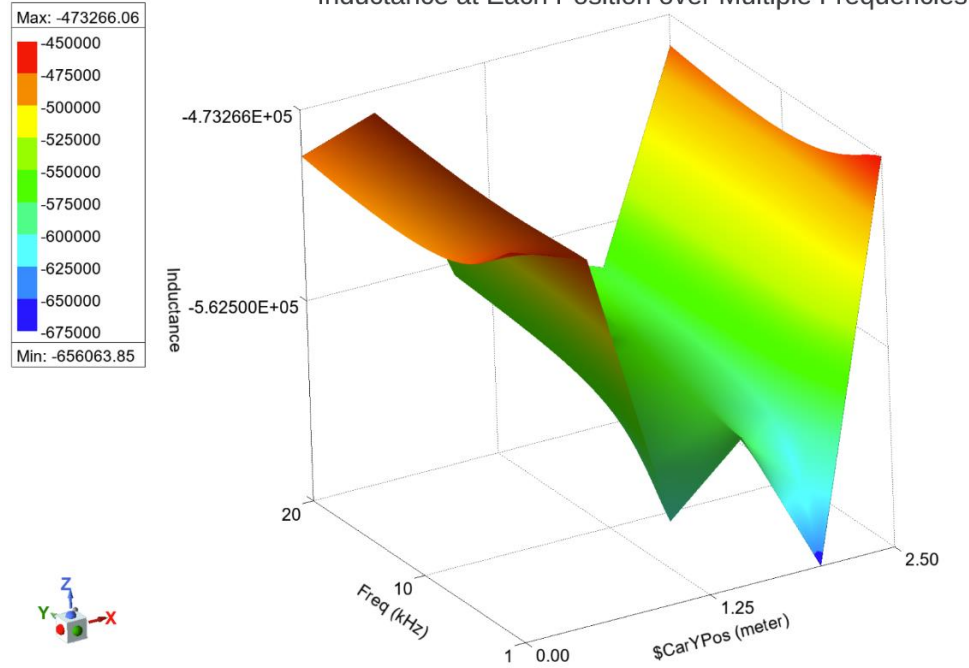


Figure 7: Graph of the Inductance of the Loop over Multiple Frequencies

2. Balancing Settings

The computational cost of simulation is very high, with some taking many hours and even multiple days. We do not have this time. In order to ensure that our simulations ran in at most an hour while maintaining accuracy, we made sure to adjust the meshing and solving settings to be fairly balanced. We tested high speed settings and found they were less accurate and less consistent than the balanced settings.

3. Parametric Sweep

A parametric sweep is a part of the Optimetrics module in HFSS. It allows for the incremental change of chosen variables, simulating at each step. We changed the distance of the vehicle over the loop in order to simulate it passing over the loop. We used six steps as too many steps can significantly increase the computational cost. We also need to set up the calculation that would be done at each step (Eq. 8). We used a general inductance equation for HFSS. The $Z(1,1)$ is the Z-parameter in HFSS that measures the input impedance of the port.⁹

$$Inductance = \frac{im(Z(1,1))}{2 * \pi * freq} \quad (8)$$

4. Finite Element Method (FEM)

Finite Element Method is a powerful tool that allows for complex physics scenarios to be solved by a computer accurately. Finite Element refers to the discretization of the model into subsections called Elements (see Figure 8).

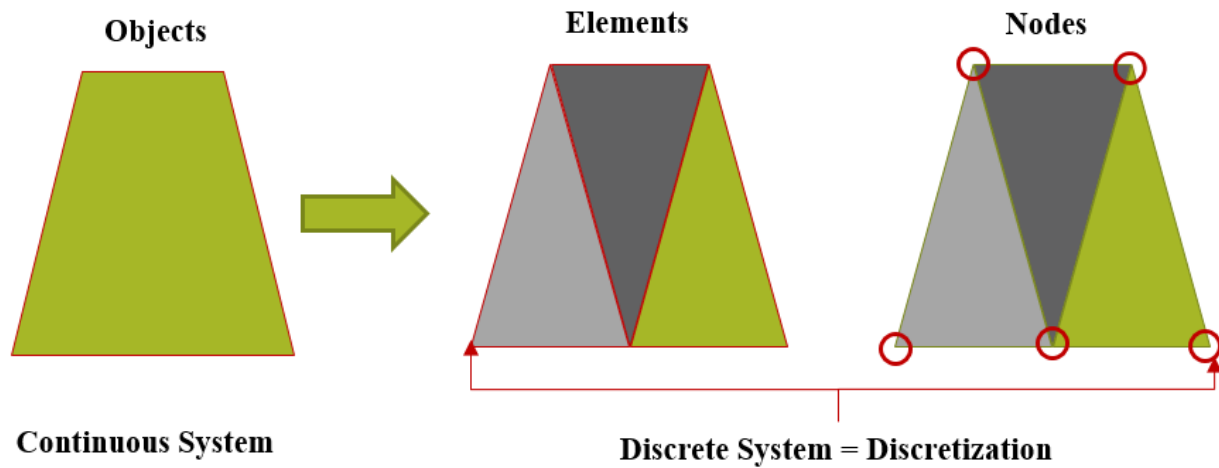


Figure 8: Finite Element Method

The solver uses elements to solve the physical scenario in the small section, and then adds the elements together to create a full solution of the model. This perturbative technique is powerful because it is easier for the computer to solve scenarios in small elements than in the entire thing. The size of the elements matters, as larger elements can cause inaccurate measurements due to low resolution. Balancing the size of elements and the runtime can be challenging. The mesh is the entire set of the elements for a model. The mesh of elements can have a variety of shapes, but we chose to use triangles. HFSS also utilizes adaptive meshing, which means the mesh is created with varied sizes of elements depending on the necessity of the element. If the element is in a region of small margin and of high importance, the element will be smaller. This adaptive meshing strategy can improve accuracy while still keeping computation cost down. FEM allows for the simulation to solve the inductance at each step (see Fig 9).

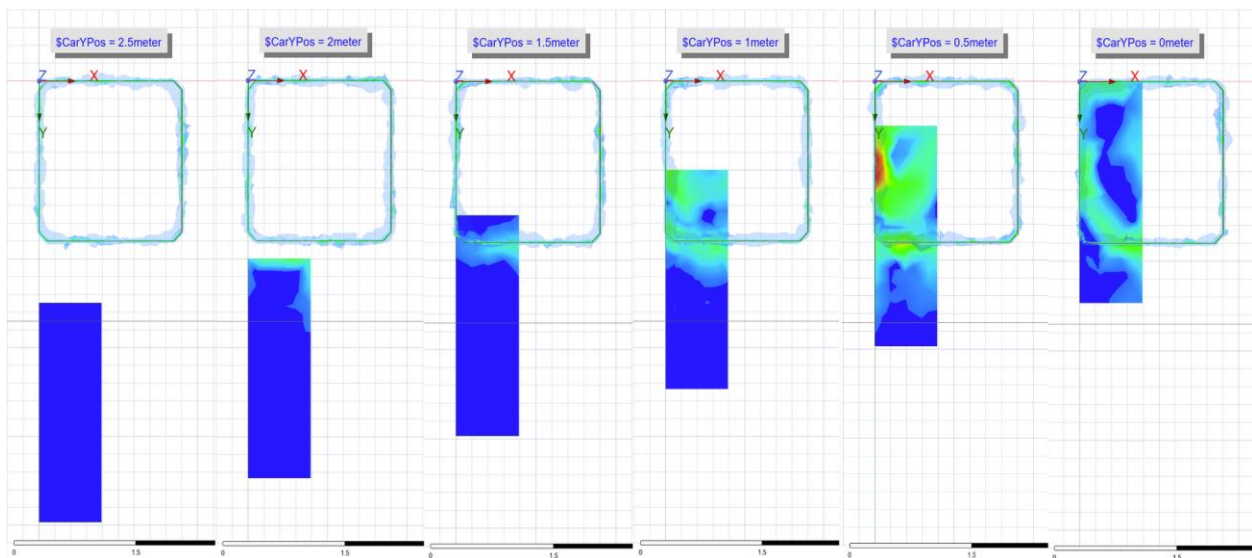


Figure 9: Position of Vehicle over the Inductive Traffic Loop

III. Machine Learning Modeling Framework

A novel hybrid machine learning framework was used that fuses physics-based simulation, deep learning, and traditional machine learning for robust vehicle position detection using inductive traffic loops (Fig. 1). The pipeline begins with high-fidelity electromagnetic simulations of inductive loop sensors generated from ANSYS HFSS with additive random noise. Each synthetic sample is labeled with a discrete class indicating four different locations in the loop.

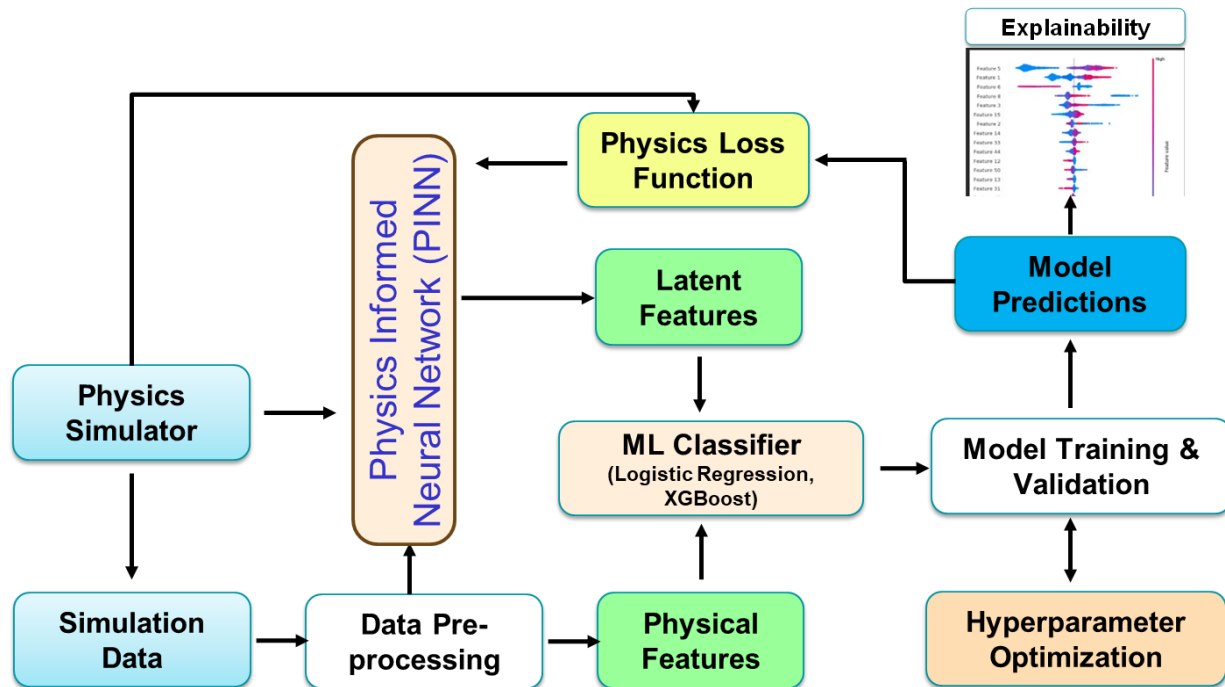


Figure 10. Hybrid machine learning framework that fuses physics-based simulation, deep learning, and traditional machine learning.

A. Software Tools and Libraries

1. Tensorflow and Keras

Tensorflow is an open-source machine learning platform that was first released by Google in 2015. Its purpose was to simplify the design, training, and complexity of computational modeling. Tensorflow provides a powerful framework for neural networks and deep learning that allow developers to create a wide variety of machine learning algorithms. ¹⁰

Tensorflow is a powerful tool that is more easily utilized through the High Level API Keras. Keras provides a lot of abstraction for the tensorflow tools that make it easier for developers to create neural networks.

The common workflow for Keras with Tensorflow is to pre-process the data, build the neural net model with Keras, and to train the model.¹¹

2. Pandas, Numpy, and Matplotlib

Pandas is a very common data science tool utilized in Python. Pandas can manipulate and control large sets of data in a tabular fashion through its Series (1D array) and DataFrame (2D array) objects. These objects allow Pandas to easily sort through and export data in CSV formats. Numpy is another common data science tool utilized in Python. Numpy provides powerful computational and mathematical tools that allow Pandas to do certain data manipulation tasks. Matplotlib is a data visualization tool that allows for the graphing of information from other tools like Pandas.

B. Data Generation

Data prediction models require large amounts of data to train. Our data generation techniques were a foundational part of creating large amounts of data even when simulations gave us little data over a long time span.

1. Original Data Set

Using a standard sized motorcycle model, we simulated it passing over the loop at 4 different positions (x_1 , x_2 , x_3 , x_4)(see Figure 10).

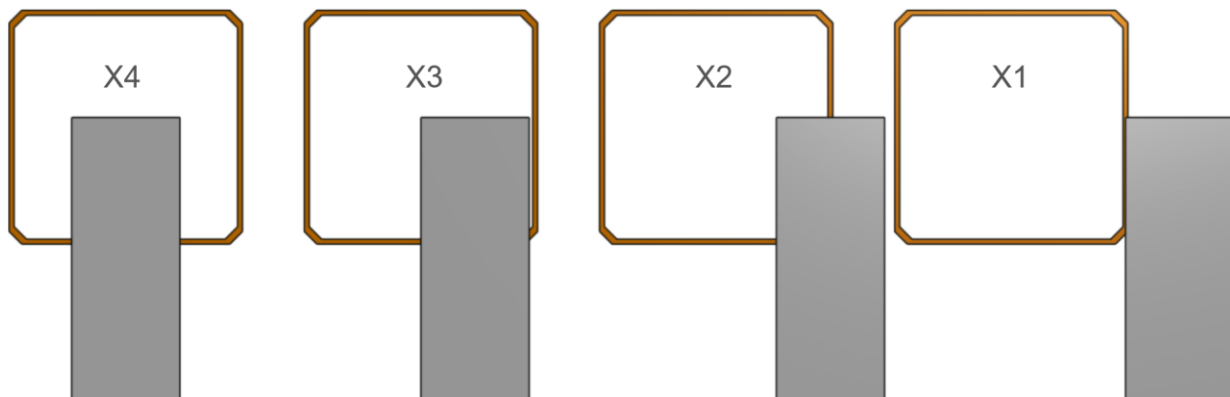


Figure 10: Motorcycle model at its 4 different positions over the inductive traffic loop

At each of the different positions, we recorded the change in inductance. The discrete change in inductance values across the different positions meant that we could generate more data from this set to train a model to determine the position of a motorcycle within the lane.

2. First Test: Gaussian Noise Distribution

Gaussian Distribution, also known as Normal Distribution, is a continuous probability distribution that takes on a bell curve shape.

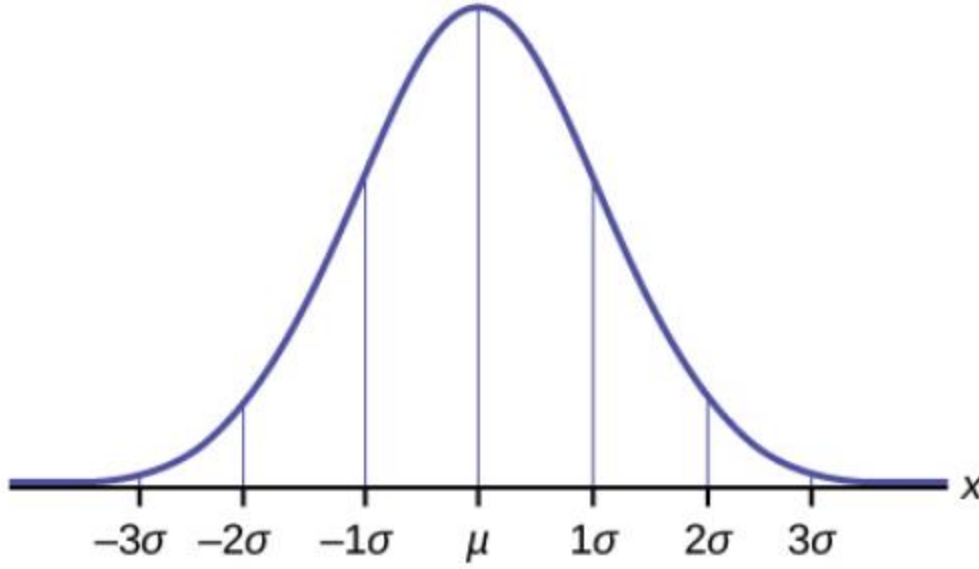


Figure 11: Gaussian distribution

Many random behaviors that occur within data can be fit to a Gaussian Distribution (see Figure 11). Since our inductance model has little data, we decided to use a technique called noise in order to fit many values to a Gaussian curve in hopes of accurately modeling the real world data that might occur at a traffic stop.

The user specifies the amount of motorcycles they want to generate (eg. 10000). We used Gaussian noise to create new random types of motorcycles and to generate new values for inductance for those motorcycles. To create the noise, we utilized Numpy's `random.normal()` function. This function uses the probability density function (PDF) for a gaussian distribution curve to calculate the probability of a certain value centered around the mean occurring (Eq. 9).¹² In our case, the mean(μ) is 1 and our standard deviation(σ) is 0.05 as we want the values to not vary heavily.

$$f(x) = \frac{1}{\sigma * \sqrt{2 * \pi}} * e^{-\frac{1}{2} * (\frac{x-\mu}{\sigma})^2} \quad (9)$$

A smaller standard deviation leads to a curve that is more concentrated around the mean. We also originally utilized a scaling factor that would scale the inductance values based on the volume of the new motorcycle compared to the old motorcycle (Eq. 10). We also tested using scaling through the area of the front face of the motorcycle, and observed better results. The new inductance was a transformation of the noise and the scale factor (Eq. 11).

$$\text{scale factor} = \frac{\text{volume}}{\text{base volume}} \quad (10)$$

$$\text{new inductance} = \text{base inductance} * \text{Gaussian noise} * \text{scale factor} \quad (11)$$

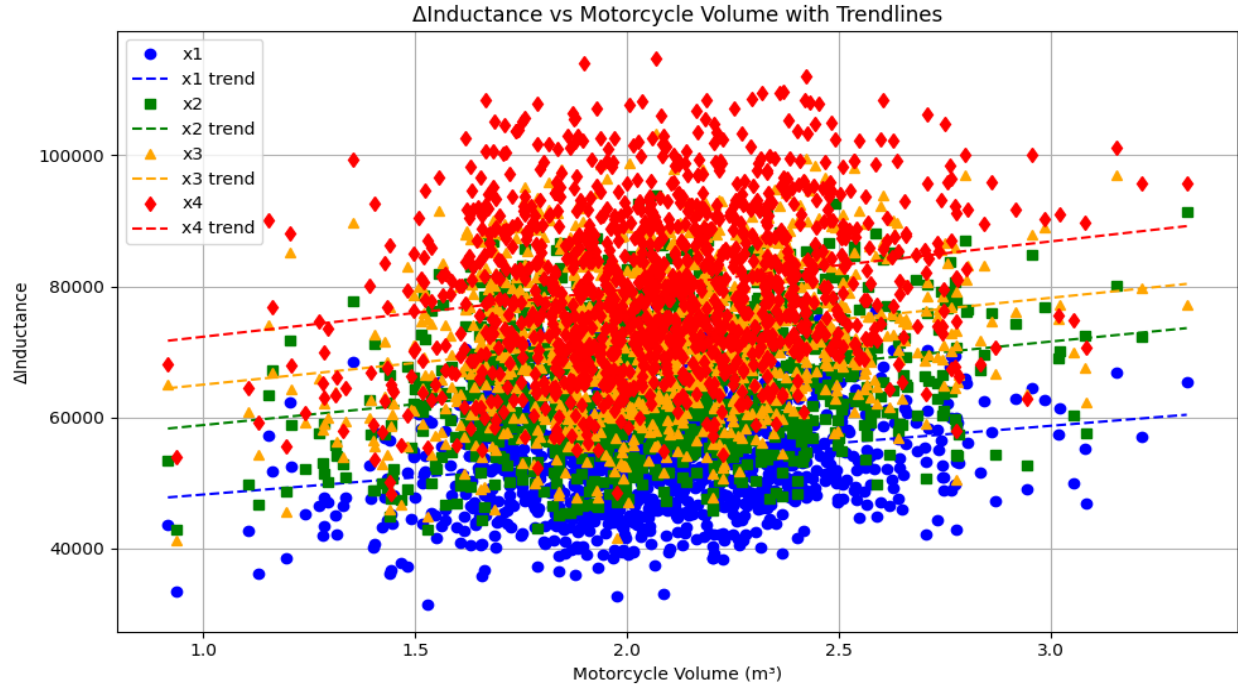


Figure 12: Change of Inductance vs Motorcycle Volume with Trendlines

We observed that with the Gaussian noise and the scaling factor, the data was too spread and would not accurately represent the real world scenario (see Figure 12). The values were scaled a lot and so spread that motorcycles in the x4 position would have inductance values of motorcycles in the x1 position, creating a continuous set of data that had non-discrete levels.

3. Second Test: Hyperbolic Secant Distribution

Hyperbolic Secant Distribution, also known as inverse-cosh distribution, is a continuous probability distribution that also takes on a bell curve shape. Its probability function is defined by sech (Eq. 13).¹³

$$f(x) = \frac{1}{2} \text{sech}\left(\frac{\pi * x}{2}\right) \quad (13)$$

The main issue we had with the Gaussian distribution was that the values were too spread out around the mean. This occurred due to a property of continuous distributions called Kurtosis. Kurtosis is the measure of the tailedness of a distribution which is linked to the frequency of extreme values.¹⁴ The Gaussian distribution is considered Mesokurtic (kurtosis = 3) and the hyperbolic secant distribution is considered Leptokurtic (kurtosis = 5 > 3)(see Figure 13). This means that the hyperbolic secant distribution has a sharper peak with less tails, or in short terms, has more values around the mean than the Gaussian distribution.

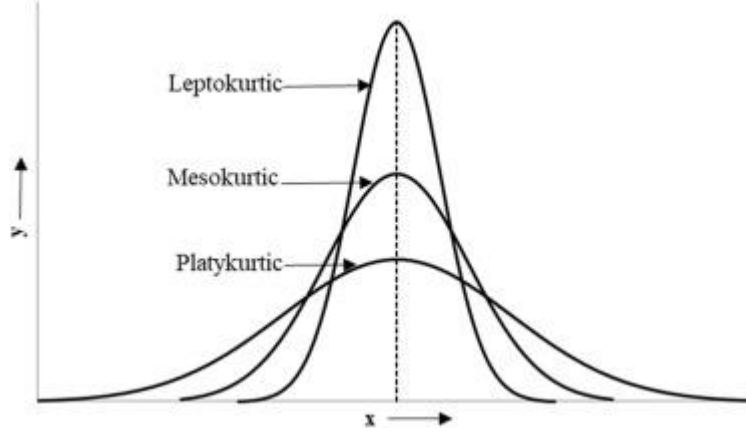


Figure 13: Kurtosis

To generate a better set of inductance values, we used noise generated with the hyperbolic secant distribution instead. Since Numpy doesn't have a sech function, we utilized a technique called inverse CDF sampling, which allows us to sample random numbers from a probability distribution. CDF stands for Continuous Distribution Function and it outputs the probability of a value X occurring in the set before the value x . The CDF of a hyperbolic secant distribution is defined with arctan and tan which Numpy does support (Eq. 14).

$$F(x) = \frac{2}{\pi} \arctan(e^{\frac{\pi * x}{2}}) \quad (14)$$

To do inverse CDF sampling, we used the inverse of the CDF function for a hyperbolic secant distribution (Eq. 15) and replaced the x in the expression with a Uniform distribution function that returns a random value from 0 to 1.¹⁵ This function represents the same distribution as the probability function.

$$X = \frac{2}{\pi} \ln(\tan(\frac{\pi * U}{2})), \quad U \in (0, 1) \quad (15)$$

By using noise generated by the hyperbolic secant distribution and a scaling factor, we had a tighter set of data that better modeled the scenario, but the scaling was still off. When we removed the scaling factor, the hyperbolic secant distribution seemed to scale the data better than with the scaling factor and was significantly better than the gaussian distribution (see Figure 14).

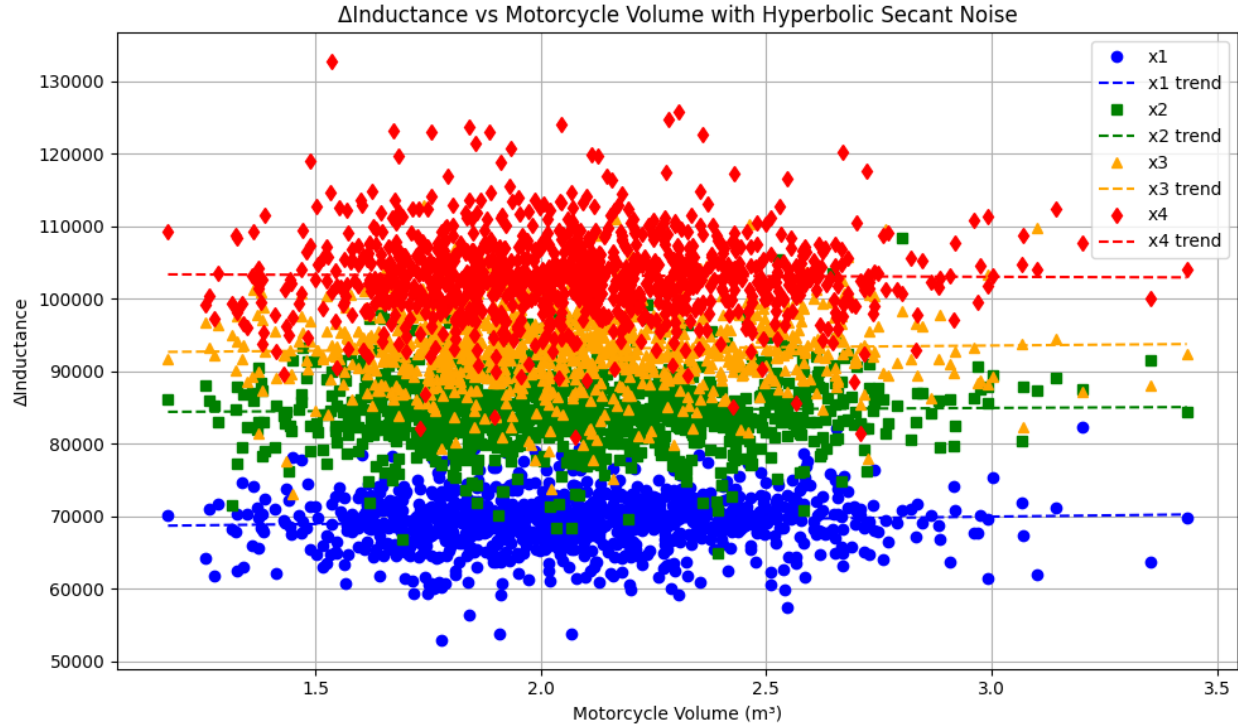


Figure 14: Change of Inductance vs Motorcycle Volume with Hyperbolic Secant Noise with Trend Lines

C. Preprocessing and Feature Scaling

Before training, all inductance values were scaled using standard normalization (zero mean and unit variance) to account for variations in magnitude across frequencies and various simulation settings. This prevents variables with larger numerical ranges from having a stronger leverage in the learning process. Labels indicating lateral motorcycle position were encoded as four categorical classes for use in classification models. Additionally, any simulated data affected by extreme noise or outliers was removed to maintain the physical validity of the dataset¹⁹. The dataset was split into training (80%) and testing (20%) sets using stratified sampling to preserve class balance.

D. PINN Training and Latent Feature Extraction

Traditional machine learning algorithms primarily leverage data-based learning processes, which can often lead to overfitting or failure to generalize, especially in scientific problems where physical laws govern the system²⁰. Physics-Informed Neural Networks (PINNs) are trained to solve supervised learning tasks while following any given laws of physics. These provide a framework to incorporate these governing equations directly into the learning process, enabling models to learn not just from data but also from the underlying physics of the system.²¹

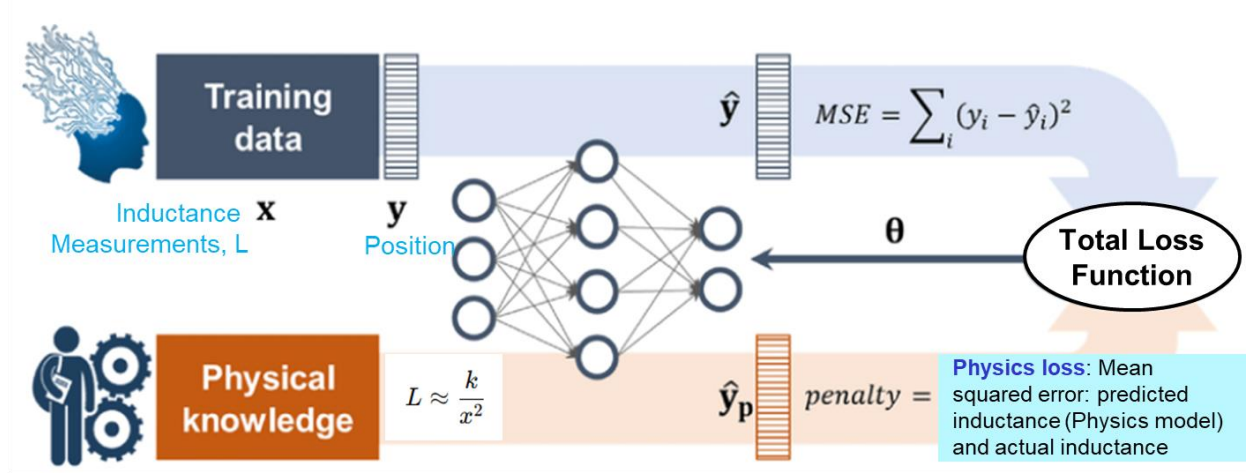


Figure 15: PINN Model learns a structure informed by real-world physical constraints. Latent features are more meaningful because they tie back to interpretable physics. ¹⁶

Figure 15 describes the loss function in the PINN consisting of two components: a data-driven term and a physics-based regularization term.¹⁶ The total loss was computed as the sum of the two losses: MSE handles the data discrepancy between the predicted and actual positions, offering robustness to outliers.²¹ The second term, the physics loss, was derived from the inverse-square relationship of inductance with position. The PINN was penalized for predicting positions that resulted in inductance values inconsistent with this physical law. This structure guides the training of the network within the constraints of the physical laws.

A Physics-Informed Neural Network (PINN) is trained on the scaled inductance inputs to predict continuous position values.²¹ Once trained, the final hidden layer's output (a vector of latent features) is extracted and used as an additional set of features for downstream machine learning models. These latent features represent the physics-based insights related to the system behavior.

E. Physical Features

To bridge the gap between raw data and position classification, several interpretable features were defined

1. Physics-Based Position Estimate

Position value is approximated based on a first-principles relationship:

$$x_{phys} = \sqrt{\frac{k_{opt}}{L}} \quad (16)$$

2. Relative Error Between NN and Physics

Based on a well-documented method, this feature defines the divergence between the PINN and the physics-based estimate.

$$\text{relative error} = \frac{x_{NN} - x_{phys}}{x_{phys} + \epsilon} \quad (17)$$

3. Implied 'k' per Sample

This is an inverse relationship, defining a governing parameter in the physical model on a per-instance basis based on model prediction, x_{NN} and observations, L :

$$k_{pred} = L * x_{NN}^2 \quad (18)$$

These features captured meaningful structure and provided an explainable relationship between observed inductance and estimated positions.

F. Feature Fusion

The PINN's latent layer outputs (dimension = 16) were concatenated (combined) with physical features (vehicle size, loop geometry, material properties) to form a richer feature space. The latent layers represent the internalization of the data for the model to understand. The model uses learned representations and known physical characteristics to make better decisions. This expanded set of features provides a generalizable predictor set, especially in regions of the input space where the raw inductance values alone were ambiguous.

G. Classification Model Training and Validation

The final state of the framework involves a multiclass classification task. The objective is to categorize a sample into one of four discrete lateral positions. Two classifiers were evaluated:

1. Logistic Regression

Logistic Regression Served as a baseline model due to its interpretability. Hyperparameters such as regularization strength (L2) and solver type were optimized. It is a classification algorithm, so it takes the inputs and transforms them to have weights, uses a function to range them from 0 to 1, and outputs the probability of the inputs being a part of a certain class. Figure 16 describes the architecture of the logistic regression model.²²

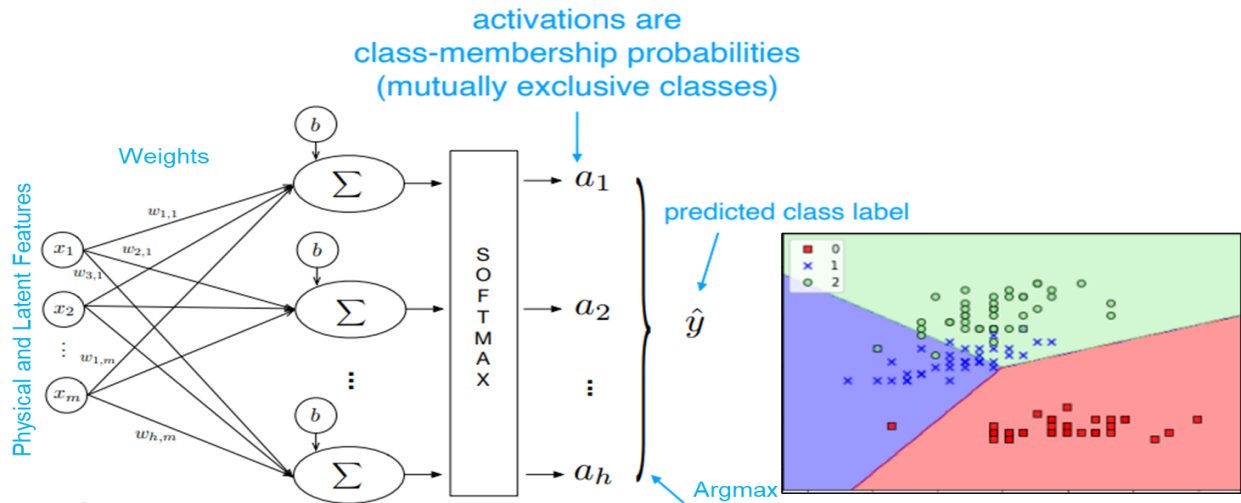


Figure 16. Multinomial Logistic Regression Architecture

1. XGBoost Classifier

Selected for its ability to model complex non-linear relationships based on ensemble modeling.²³ XGBoost will make a bunch of sub-groups called bootstrap samples and will organize these into Classifiers. The number of classifiers can be edited as a hyperparameter. Each classifier is trained independently and then combined together into the ensemble classifier. The classifier will take the outputs of all classifiers in order to make the best prediction with the test data. Figure 17 describes the model architecture and key hyperparameters including, learning rate and number of estimators to be optimized for optimal model selection.

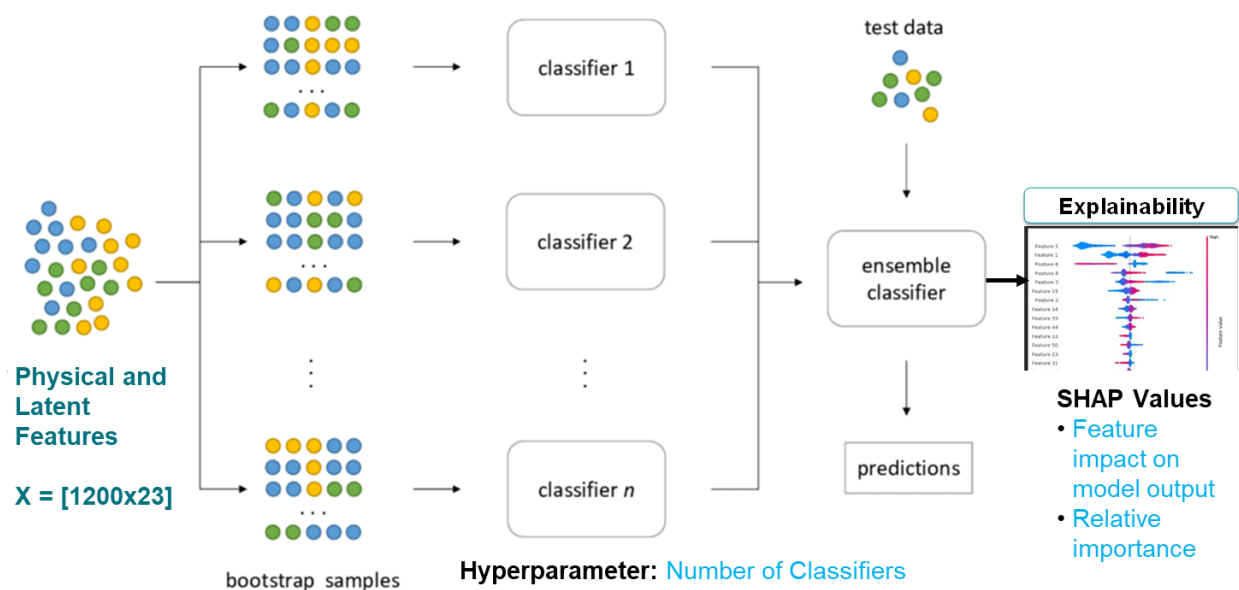


Figure 17: XGBoost Classifier Architecture¹⁷

These models are trained using three feature combinations:

1. Inductance Only
2. Inductance + Physics-Based Features
3. Inductance + Physics + Latent Features

Performance metrics such as accuracy, and ROC (Receiver Operating Characteristic) curves are used to evaluate each combination listed above.¹⁹ Both classifiers were trained using k-fold validation and evaluated based on accuracy of the test data as defined in figure 18.

The best results are consistently achieved with the hybrid feature set, showing the value of integrating physically meaningful representation into machine learning workflows.

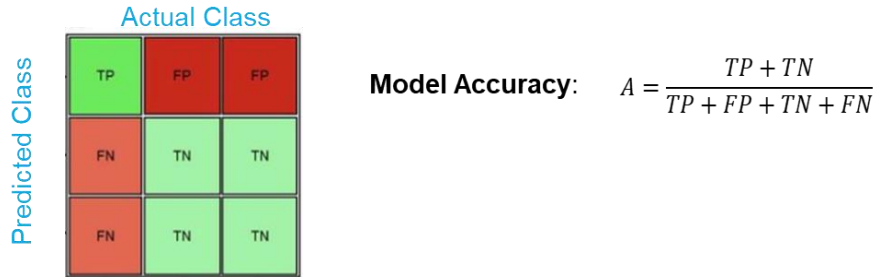


Figure 18: contingency Matric for three class problem¹⁸

H. Results and Analysis

1. Model Performance Metrics

The XGBoost model with both physical and latent features outperformed all other models. Notably, the accuracy gain was substantial when moving from raw inductance to the hybrid feature set. Demonstrating the utility of combining underlying physics knowledge with data-driven learning (Figure 19).

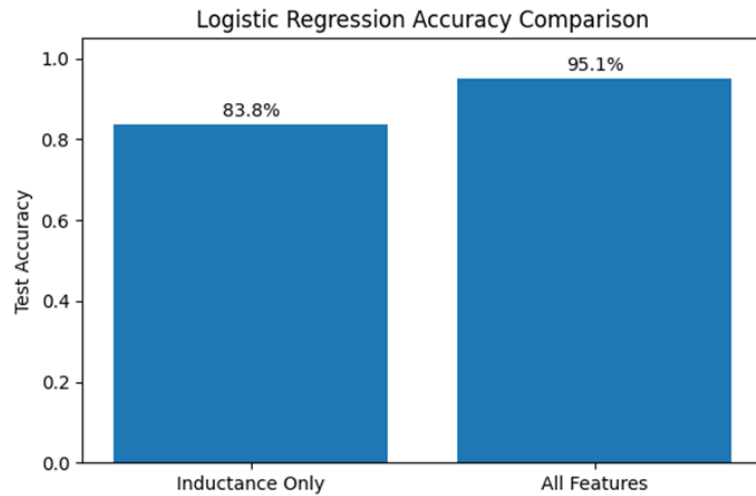


Figure 19. Logistic Regression Accuracy Comparison

1. ROC Curve and Feature Importance

A macro-averaged ROC curve was plotted for the classification model. It is a graphical method used to evaluate the performance of a classification model depicting the trade-off between the True-Positives (TP) and the False Positives (FP) rates at various threshold settings.²⁴ Key output is Area Under Curve (AUC), a value of 1.0 means perfect model and a value of 0.5 means random guessing.¹⁹ In our study, the AUC improved significantly when using the enhanced feature set, confirming higher confidence in classification predictions (Figure 20)

Feature importance analysis through XGBoost's gain metric revealed:

1. The first latent feature from the PINN had the highest importance
2. The scaled inductances followed
3. The difference in K values was the third largest contributing factor towards the model's prediction

The ranking further substantiates the hypothesis that physics-informed and abstracted latent representations offer superior predictive power over raw signals alone (Figure 21).

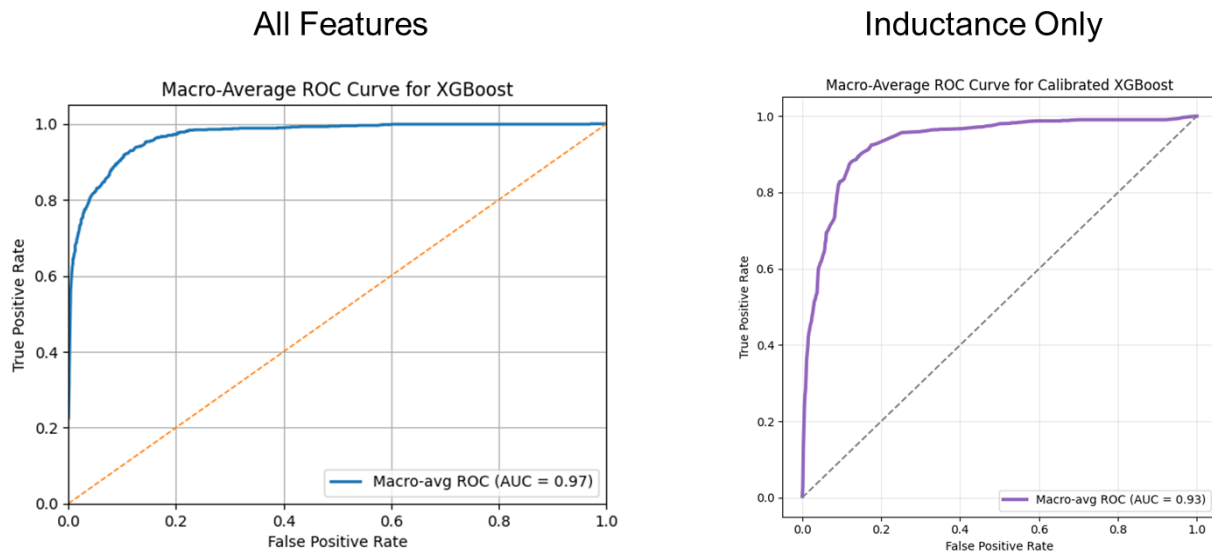


Figure 20. XGBoost ROC Curve Comparison

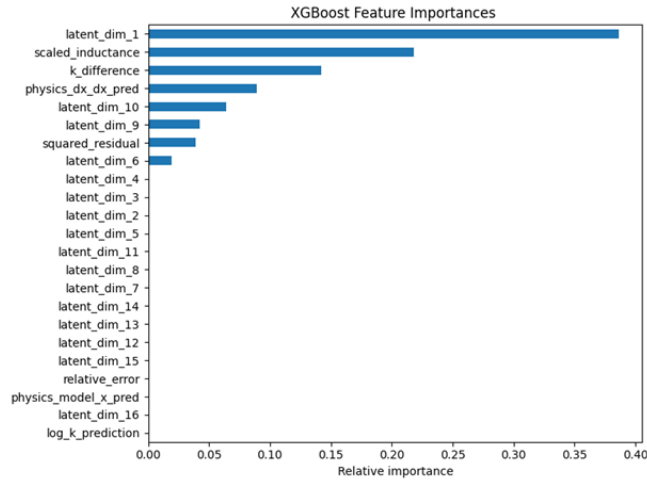


Figure 21. Feature Importance Map for XGBoost Model

III. Discussion

This study demonstrates the effectiveness of integrating physics-based understanding with machine learning for vehicle position classification. The use of PNNs allows for insightful, more robust feature extraction, particularly when dealing with noisy data, common in embedded transportation systems.

We concluded that the goal of accurately determining the position of the motorcycle was achieved, and the model with the additional features such as the Physical Features + PINN features outperformed the traditional linear regression model. Also, we concluded that the first feature from the PINN was more important to the classification than the induction was. Lastly, we concluded that the Hyperbolic Secant Noise was better at modeling data for the system than Gaussian Noise due to its tighter distribution.

Based on the outcomes of the testing, it can be concluded that a real world model of an inductive traffic loop can accurately detect the position of a motorcycle within a lane using a Physics Informed Neural Network (PINN).

IV. Future Research

A. More Data

We would have liked to gather more simulation data to make the model more accurate. If many different scenarios could have been fed into the network, it would have the chance of making better predictions. This would also mean we could use less generated data that fits better to the simulation data.

B. Vehicle Detection Feature

Using our physics based model, we could possibly detect different types of vehicles that pass over the loop. Since there is a physical difference between vehicles, they could be detected and classified as a certain type as they pass over the loop, hence giving more traffic data to a controller.

C. Speed and Acceleration

The data of inductance, if measured at more steps, could be used to track the speed of a vehicle passing over it. By going through the change in inductance through each time step, the positioning and speed could be calculated through a function. A model could also be used to predict the speeds based on trained sets of known speeds and inductance values associated with those speeds.

D. Optimizing Geometry

Depending on the geometry of the loop, different pieces of data can be acquired. In order to design loops that best fit the desired data collection, optimization techniques could be used. Optimization algorithms can edit the parameters based on certain constraints to reach an optimal goal. These algorithms have powerful implications for inductive loop design as the loops could become multifunctional and more applicable to many areas of traffic theory.

E. X and Y Position Detection

For this project, we detect the x position of a motorcycle passing over the loop, but in theory, the y position could also be calculated. The simulations would need to involve many steps in order to get a high resolution view of vehicle movement. The data extracted from these simulations could contribute to a sort of field or matrix that represents that vehicle's data at different x and y positions over the loop. A model could also be trained on these different fields and eventually would be able to decide where a vehicle is based on its data at a certain position.

V. References

1. Klein, L. A., Mills, M. K., & Gibson, D. R. P. (2006). *Traffic detector handbook: Third edition—Volume I* (Report No. FHWA-HRT-06-108). U.S. Department of Transportation, Federal Highway Administration. <https://www.fhwa.dot.gov/publications/research/operations/its/06108/02.cfm>
2. Liao, S., Dourmashkin, P., & Belcher, J. W. (2004). *Faraday's law of induction* (Physics 8.02 TEAL Course Notes, Chapter 10) [PDF]. Massachusetts Institute of Technology. <http://web.mit.edu/8.02t/www/802TEAL3D/visualizations/coursenotes/modules/guide10.pdf>
3. Byju's. (n.d.). *Magnetic flux*. BYJU'S – The Learning App. Retrieved July 24, 2025, from <https://byjus.com/physics/magnetic-flux/>
4. OpenStax. (2016). 23.5: *Faraday's law of induction—Lenz's law*. In *College Physics* (1st ed.). LibreTexts. [https://phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_\(OpenStax\)/23%3A A Electromagnetic Induction AC Circuits and Electrical Technologies/23.05%3A Faradays Law of Induction- Lenzs Law](https://phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_(OpenStax)/23%3A_A_Electromagnetic_Induction_AC_Circuits_and_Electrical_Technologies/23.05%3A_Faradays_Law_of_Induction-_Lenzs_Law)

5. Byju's. (n.d.). *Electromagnetic induction*. BYJU'S – The Learning App. Retrieved July 24, 2025, from <https://byjus.com/physics/induction/>
6. DoITPoMS. (n.d.). *Ferromagnetic materials*. Department of Materials Science and Metallurgy, University of Cambridge. Retrieved July 24, 2025, from <https://www.doitpoms.ac.uk/tlplib/ferromagnetic/printall.php>
7. Kerner, B. S. (2004). Three-phase traffic theory and highway capacity. *Physica A: Statistical Mechanics and Its Applications*, 333, 379–440. <https://doi.org/10.1016/j.physa.2003.10.017>
8. Ansys Inc. (2020). *Lesson 6: Port basics* (HFSS Getting Started 2020 R2) [PDF]. https://innovationspace.ansys.com/courses/wp-content/uploads/sites/5/2021/07/HFSS_GS_2020R2_EN_LE6_Port_Basics.pdf
9. Coil32. (n.d.). *HFSS simulation of a helix coil*. Coil32 Online Calculator. Retrieved July 24, 2025, from <https://coil32.net/theory/hfss-helix-coil.html>
10. NVIDIA. (n.d.). *TensorFlow*. NVIDIA Glossary. Retrieved July 24, 2025, from <https://www.nvidia.com/en-us/glossary/tensorflow/>
11. Keras. (n.d.). *About Keras*. Keras.io. Retrieved July 24, 2025, from https://keras.io/getting_started/about/
12. Illowsky, B., & Dean, S. (2018). *Introductory statistics*. OpenStax, Rice University.
13. Siegrist, K. (n.d.). 5.31: *The hyperbolic secant distribution*. In *Probability, mathematical statistics, and stochastic processes*. LibreTexts. Retrieved July 24, 2025, from [https://stats.libretexts.org/Bookshelves/Probability_Theory/Probability_Mathematical_Statistics_and_Stochastic_Processes_\(Siegrist\)/05:_Special_Distributions/5.31:_The_Hyperbolic_Secant_Distribution](https://stats.libretexts.org/Bookshelves/Probability_Theory/Probability_Mathematical_Statistics_and_Stochastic_Processes_(Siegrist)/05:_Special_Distributions/5.31:_The_Hyperbolic_Secant_Distribution)
14. Navarro, D. (n.d.). 5.03: *Skew and kurtosis*. In *Learning statistics with R: A tutorial for psychology students and other beginners*. LibreTexts. Retrieved July 24, 2025, from [https://stats.libretexts.org/Bookshelves/Applied_Statistics/Learning_Statistics_with_R_-_A_tutorial_for_Psychology_Students_and_other_Beginners_\(Navarro\)/05%3A_Descriptive_Statistics/5.03%3A_Skew_and_Kurtosis](https://stats.libretexts.org/Bookshelves/Applied_Statistics/Learning_Statistics_with_R_-_A_tutorial_for_Psychology_Students_and_other_Beginners_(Navarro)/05%3A_Descriptive_Statistics/5.03%3A_Skew_and_Kurtosis)
15. Fischer, M. (2004). Skew generalized secant hyperbolic distributions: Unconditional conditional fit to asset returns. *Austrian Journal of Statistics*, 33, 183–196.
16. Kim, Sungwoo, et al. "Data-Driven Prognostics with Low-Fidelity Physical Information for Digital Twin: Physics-Informed Neural Network." *Structural and Multidisciplinary Optimization*, vol. 65, no. 4, 2022. <https://doi.org/10.1007/s00158-022-03348-0>
17. Hachcham, A., XGBoost: Everything You Need to Know, 2025, <https://neptune.ai/blog/xgboost-everything-you-need-to-know>
18. Abraham, A. et al. (2023). Comparative Analysis of Machine Learning Models for Image Detection of Colonic Polyps versus Resected Polyps. 10.20944/preprints202308.0715.v1
19. Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. 2nd ed., O'Reilly Media, 2019.
20. Karniadakis, George Em, et al. "Physics-Informed Machine Learning." *Nature Reviews Physics*, vol. 3, 2021, pp. 422–440. <https://doi.org/10.1038/s42254-021-00314-5>

21. Raissi, Maziar, Paris Perdikaris, and George Em Karniadakis. "Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations." *Journal of Computational Physics*, vol. 378, 2019, pp. 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>.
22. Raschka, Sebastian. "Multinomial Logistic Regression." *Intro to Deep Learning*, 2021, <https://sebastianraschka.com/blog/2021/dl-course.html>
23. Chen, Tianqi, and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>
24. Fawcett, Tom. "An Introduction to ROC Analysis." *Pattern Recognition Letters*, vol. 27, no. 8, 2006, pp. 861–874. <https://doi.org/10.1016/j.patrec.2005.10.010>.